SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	AI&ML	**Date:**	06/08/2026

Code of Conduct

I __A. Mounish / 192472273__ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System

Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:
"Data science is powerful, data science drives innovation, data science is evolving."
The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.
The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ = 0.5 (Trigram)
- λ_{123} = 0.3 (Bigram)
- λ = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.

3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.
Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging. Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.
The sentence processed by the system is:
"Economic growth increases employment." Initial
POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:
Change NNS to VBZ if the preceding word is tagged as NN.
The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
   - economic (120 occurrences)
   - growth (450 occurrences)
   - increases (210 occurrences)
   - employment (380 occurrences)

   Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.
4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

CASE STUDY 1 – Smart Mobile Keyboard Prediction System

Aim: To analyze Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy for next-word prediction in a smart keyboard.

Q1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability . Interpret its role.

Formula

Given

-
-

Calculation

Answer

Bigram Probability = 1.0

Interpretation

A probability of **1.0 (100%)** means that every occurrence of the word **"data"** in the corpus is followed by **"science"**. Therefore, the keyboard confidently predicts **science** whenever the user types **data**, improving prediction accuracy and typing speed.

Q2. Apply the Backoff Model for the unseen sequence "data science improves".

Solution

The trigram **"data science improves"** does **not appear** in the training corpus.

Using the **Backoff Model**:

1. Check Trigram → **Not Found**
2. Backoff to Bigram → **science improves** → Not Found
3. Backoff to Unigram → **improves**

Since only the unigram is available,

Answer

The probability is estimated using the **lower-order unigram probability**.

Explanation

The Backoff Model avoids assigning zero probability to unseen word sequences. This enables predictive keyboards to continue suggesting reasonable words even when users type new or rare phrases, making real-time prediction more robust.

Q3. Compute Deleted Interpolation Probability for "data science is".

Formula

Given

-
-
-

Using the corpus:

- Trigram probability = 1
- Bigram probability = 1
- Unigram probability = 1

Calculation
Answer
Deleted Interpolation Probability = 1.0
Discussion
Deleted Interpolation combines trigram, bigram, and unigram probabilities. This balances high accuracy from higher-order models with better coverage from lower-order models, reducing problems caused by sparse data.

Q4. Calculate the Entropy.

Given

-
-

Formula
Calculation
Answer
Entropy = 0.92 bits
Interpretation
The entropy is relatively low, indicating **less uncertainty** in predicting the next word. Since **"is"** has a much higher probability than **"drives"**, the keyboard predicts **"is"** with greater confidence, improving prediction reliability and user experience.

Case Study 1 Result
The Bigram, Backoff, Deleted Interpolation, and Entropy techniques together improve next-word prediction accuracy by handling both common and unseen word sequences efficiently in AI-powered mobile keyboards.

CASE STUDY 2 – AI-Powered Customer Support Chatbot

Aim: To analyze Part-of-Speech (POS) tagging using the Penn Treebank Tag set and Hidden Markov Model (HMM) for resolving lexical ambiguity in chatbot systems.

Q1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book".

Sentence 1
Book a flight ticket now.

Word	POS Tag	Meaning
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Sentence 2

This book is interesting.

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Justification

- In **"Book a flight ticket now."**, **Book** is an action, so it is tagged as **VB (Verb)**.
- In **"This book is interesting."**, **book** refers to an object, so it is tagged as **NN (Noun)**.

Thus, the POS tag depends on the context of the sentence.

Q2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB).

Given

-
-

Formula

Calculation

Answer

Likelihood of "book" as Verb = 0.30

Explanation

Since the sentence begins with **"Book"**, which is an imperative command, the HMM assigns a higher probability to the **Verb (VB)** tag. Therefore, the chatbot correctly interprets **Book** as an action rather than a noun.

Q3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging.

Rule-Based Tagging	HMM Tagging
Uses predefined grammar rules	Uses probabilities from training data
Simple to implement	Learns from large corpora
Less effective for ambiguous words	Handles ambiguity effectively
Lower accuracy on unseen sentences	Higher accuracy for real-world text
Difficult to maintain large rule sets	Easily scalable

Recommendation

Stochastic (HMM) Tagging is recommended for large-scale chatbot deployment because it uses contextual probabilities, resolves lexical ambiguity more accurately, adapts to diverse user inputs, and provides better overall tagging performance than rule-based methods.

Q4. Evaluate the role of standardized POS tagsets and word classes in chatbot performance.

Answer

Standardized POS tagsets such as the **Penn Treebank Tagset** improve chatbot performance by:

- Providing consistent grammatical classification of words.
- Improving intent detection through accurate identification of nouns, verbs, adjectives, and adverbs.
- Enhancing response generation by understanding sentence structure.
- Resolving contextual ambiguity in user queries.
- Increasing the overall accuracy and reliability of NLP-based chatbot systems.

Thus, standardized POS tagging enables chatbots to understand natural language more effectively and generate contextually appropriate responses.

Case Study 2 Result

The Penn Treebank POS Tagset and Hidden Markov Model successfully resolve lexical ambiguity by assigning context-aware POS tags. This improves intent detection, response generation, and the overall effectiveness of AI-powered customer support chatbots.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________